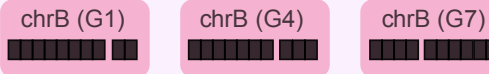
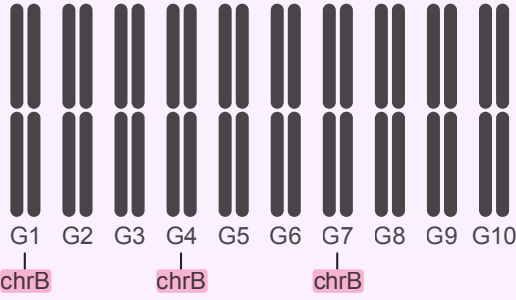


1 Assemble B chromosomes (chrB) from reference genomes

10 high-quality *Distichlis* genome assemblies



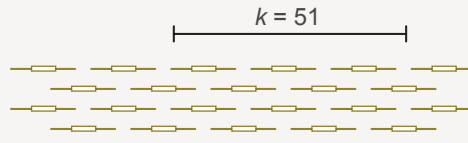
QC and curation

- Remove Ns and gaps
- Retain longest chrB contigs
- Orient and standardize
- Annotate the repeat regions and telomere

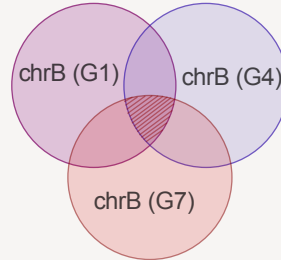
2 Build candidate chrB k-mers

Convert chrB sequences to k-mers ($k = 51$) with **Jellyfish**

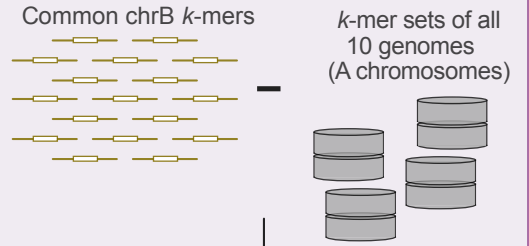
...ATGCTAGCTAGCTACGATCGA...



Intersect k-mers across the 3 chrB assemblies

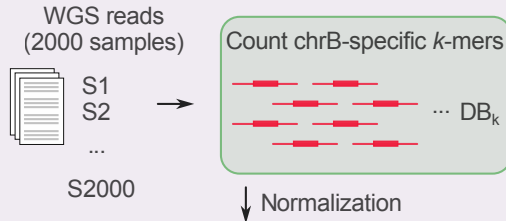


3 Remove k-mers present in A Chromosomes (10 genomes)



Output: chrB-specific k-mer database
(High-confidence, unique to chrB)
DB_k ($|\text{DB}_k| = M$ k-mers)

4 Screen 2,000 WGS samples



$$K_i = \frac{C_i}{N_i - k + 1} \times 10^9$$

Where:

C_i = number of chrB-specific k-mer hits in sample i
 N_i = total number of bases in sample i
 k = k-mer size (51 bp)

Classification

Predict chrB copy number

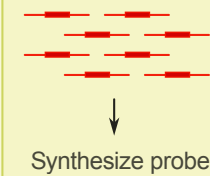
$$\hat{B}_i = \frac{K_i - \mu_o}{\mu_i - \mu_o}$$

Where:
 μ_o = mean K_i of 0B (no B) samples
 μ_i = mean K_i of 0B (no B) samples

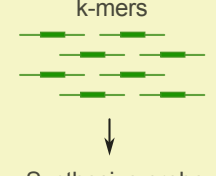
$\hat{B}_i < 0.5 \rightarrow 0B$
 $0.5 \leq \hat{B}_i < 1.5 \rightarrow 1B$
 $1.5 \leq \hat{B}_i < 2.5 \rightarrow 2B$
 ...

5 Experimental validation by FISH (cytology)

Probe design
chrB-specific k-mers



Probe design
Centromere-specific k-mers



FISH on root tip / meiotic cells

